

# Spencer Levine

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Mechanical engineering student seeking hands-on experience with the fabrication of robots, and complex mechanical assemblies and part design. Long-term goal: developing robots for lunar resource extraction. Outside of engineering: skiing, playing guitar, political science, space, and [I can backflip](#).

## Skills

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**CAD:** Fusion, SolidWorks, Onshape, AutoCAD, 3ds Max, Revit

**Computer:** Java, Python, C++, JavaScript

**Machines:** Lathe, Mill, Sandblaster, Basic CNC operation, 3D printing

## Relevant Work

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### UTX Inc.

Holmes, New York

Intern

May 2026 - Present

- Design products to streamline the inspection and production of transducers to increase efficiency by 15%
- Convert 2D drawings from pen and paper and AutoCAD into 3D models using Fusion
- Manufacture high-tolerance transducers to inspect the integrity of materials

### Nemitz Robotics Group

Medford, Massachusetts

Intern

April 2026 - Present

- Designing a [robotic gripper](#) to autonomously assemble crab robots that will disable mines in former war zones

### John Jay Robotics

Cross River, New York

FIRST Robotics Mentor

Mar 2024 - Present

- John Jay Robotics alumni; Co-led team, made it to FTC NY State Championship and World Championship
- Mentor four robotics teams of high school students competing in the FIRST Tech Challenge (FTC), including one that qualified for the FTC World Championship
- Engage 35+ non-engineering-background students in skills such as CAD, Java coding, and basic robot design

### Elaine Short Human Robot Interaction Lab

Medford, Massachusetts

Research Assistant

Jan 2026 - April 2026

- Designed a low-cost [robotic gripper](#) using 3D printing, 25 times cheaper than commercially available grippers
- Supported PhD research by designing components for robotic environments and model robotic arms

### Giuliante Machine Tool Inc

Peekskill, New York

Intern

May 2025 - Aug 2025

- Modeled and created operation sheets for 20+ parts in SolidWorks to be later machined and sold to 10+ clients
- Operated various machines, including CNCs, shapers, and bandsaws, to make gears and shafts

## Prominent Projects

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### [Robotic Arm](#)

Jul 2025 - Aug 2025

- Designed and built a robotic arm to explore different mechanical linkages, such as 4-bars and chain-driven components in order to learn how these different mechanisms affect motion and mechanical design tradeoffs
- Built a full CAD assembly of the robot in Fusion, then 3D printed and assembled

### [Swerve Drive Robot](#)

Feb 2025 - Apr 2025

- Built a fully 3D-printed swerve-drive robot with custom gear mechanisms as a challenge for myself to learn more about 3D printing and its capabilities, including the use of TPU filament, and to learn about compound gears
- Fabricated a full CAD assembly in Fusion, coded the robot in Arduino and Python, integrating wireless game controller teleoperation

## Education

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### Tufts University School of Engineering

Expected Graduation: May, 2028

*Bachelor of Science in Mechanical Engineering*

GPA: 3.67 / 4, Dean's List

Relevant Coursework: Computer-Aided Design, Materials & Manufacturing I, Mechanics I & II, Engineering Design I, Thermal Fluid Systems I, Ethics of Product Design, Energy Storage Systems

### John Jay High School

Graduated 2024